English translation prepared with ChatGPT (OpenAI, GPT-6), 26 October 2026, for translation of the Bulgarian text into English and LaTeX formatting. The Bulgarian originals remain the reference; translation may contain errors.

English transcription of the original handwritten solution outline. Its calculations and incomplete final answers are retained; apparent source slips are identified in notes.

Problem 1

Tourists form 2/3 of the visitors and answer correctly with probability 1/6; locals form 1/3 and answer correctly with probability 1. Let L and R denote the events that left and right, respectively, are the correct directions. By Bayes' formula,

$$\begin{aligned}\mathbb{P}(L \mid \text{answer } L) &= \frac{\mathbb{P}(\text{answer } L \mid L)\mathbb{P}(L)}{\mathbb{P}(\text{answer } L)} \\ &= \frac{(\frac{2}{3}\cdot\frac{1}{6}+\frac{1}{3})\cdot\frac{1}{2}}{\mathbb{P}(\text{answer } L \mid L)\mathbb{P}(L)+\mathbb{P}(\text{answer } L \mid R)\mathbb{P}(R)} \\ &= \frac{4/9}{4/9+\frac{2}{3}\cdot\frac{5}{6}} = \frac{4}{9}.\end{aligned}$$

Similarly,

$$\begin{aligned}\mathbb{P}(L \mid \text{answers } LL) &= \frac{\frac{2}{3}(\frac{1}{6})^2+\frac{1}{3}}{\frac{2}{3}(\frac{1}{6})^2+\frac{1}{3}+\frac{2}{3}(\frac{5}{6})^2} \\ &= \frac{2+36}{38+50} = \frac{38}{88} = \frac{19}{44}.\end{aligned}$$

Problem 2

Model bit i, $i = 1, \ldots, 8$, by a Bernoulli random variable X_i with probability 2^{-i} of being 1. The expected number of 1-bits is

$$\mathbb{E}(X_1 + \cdots + X_8) = \frac{1}{2} + \frac{1}{4} + \cdots + \frac{1}{2^8} = \frac{255}{256}.$$

For the expected numerical value,

$$\begin{aligned}\mathbb{E}[X_1\cdot 2^7 + X_2\cdot 2^6 + \cdots + X_8\cdot 2^0] &= \frac{1}{2}\cdot 2^7 + \frac{1}{4}\cdot 2^6 + \cdots + \frac{1}{2^8}\cdot 1 \\ &= 2^6 + 2^4 + 2^2 + 2^0 + 2^{-2} + 2^{-4} + 2^{-6} + 2^{-8}.\end{aligned}$$

The original's reported numerical answer is 85.328125. See Note 1.

Problem 3

Recall that $f_U(t) = \mathbf{1}_{[0,1]}(t)$ and $\mathbb{P}(U > 0.75) = 0.25 =: p$.

Method I

The number N of observations greater than 0.75 is $\text{Bin}(10, p)$, so

$$\mathbb{E}N = 10p = 2.5, \qquad \text{Var}(N) = 10p(1-p) = 1.875.$$

The estimate obtained by A and B is the proportion $N/10$. Hence

$$\mathbb{E}(N/10) = \frac{\mathbb{E}N}{10} = 0.25, \qquad \text{Var}(N/10) = \frac{1.875}{100}.$$

Method II

Here the estimate is $\widetilde{X} = \mathrm{Bin}(1000, p)/1000$. Thus

$$\mathbb{E}\widetilde{X} = 0.25, \qquad \mathrm{Var}(\widetilde{X}) = \frac{10^3 p(1-p)}{10^6} = \frac{1.875}{10^4}.$$

3. Rounds

$$\mathbb{P}(\text{both numbers} > 0.75) = \left(\frac{1}{4}\right)^2 = \frac{1}{16}.$$

By the strong law of large numbers, this happens in a proportion 1/16 of the rounds. Also,

$$\mathbb{P}(\text{at least one number} > 0.75) = 1 - \left(\frac{3}{4}\right)^2 = \frac{7}{16},$$

so the expected number of rounds is $\mathbb{E}[\mathrm{Ge}(7/16)] = 16/7$.

Problem 4

1. The original writes

$$\begin{aligned}\mathbb{P}(N_1(-10, 10) > N_2(0, 100)) &= \mathbb{P}(N(10, 110) > 0) \\ &= \mathbb{P}\left(N(0,1) > \frac{10}{\sqrt{110}}\right) \approx \cdots . \qquad \text{(Note 2)}\end{aligned}$$

2. The final line of the original outline is

$$\mathbb{P}(N_1(-10, 10) < -0.1). \qquad \text{(Note 3)}$$

Transcription notes

1. The displayed eight-term sum is 85.33203125. The original's 85.328125 omits the final 2^{-8} contribution.
2. The intermediate normal distribution for the difference should have mean -10. The source writes $+10$, while its final standard-normal threshold is consistent with mean -10.
3. The problem measures the change in percentage points. A reduction from 250,000 to 225,000 euros is -10 percentage points, whereas the outline writes the fractional change -0.1. The outline does not finish the numerical calculations or provide the second case.